

GLASS POCKET

A Snowflake accountability engine for charitable giving.
Multi-tab Streamlit application, Solana compressed-
NFT receipt ledger, differential privacy serving layer.

Target

DEV.to Weekend Challenge
Generosity Edition

Primary category

Best Use of Snowflake

Secondary

Best Use of Solana

Deadline

Mon 07 Sep 2026
12:29 PM IST

00 / How to use this document

Operating instructions for the implementing agent. Read this section fully before writing any code.

This document is the single source of truth for the GLASSPOCKET build. It is written to be executed by an autonomous coding agent with a human operator reviewing at checkpoints. Every section is normative. Where a value is specified, use that value exactly rather than substituting a preference.

Execution rules

1. **Build in the order given in Section 09.** The order is dependency-sorted and risk-sorted. The riskiest item, the differential privacy policy, is deliberately first because it is the only component that can fail in a way that cannot be fixed later.
2. **Do not begin the Streamlit application until the data layer passes its acceptance checks.** A beautiful frontend over an empty warehouse is worth zero points in this challenge.
3. **Never call an external network service from inside Snowflake.** The account is a trial account. External network access is unavailable and will fail at compile time with error 509009. All outbound work happens in the Node bridge described in Section 08.
4. **Precompute aggressively.** Nothing heavier than a single indexed SELECT may execute during the demo. Every embedding, similarity pair, geospatial join and blockchain asset ID is materialised into a table before demo time.
5. **Label synthetic data everywhere it appears.** This is a project about fraud. Presenting fabricated evidence as real would be self-defeating and dishonest. Section 05 defines the labelling contract.
6. **Commit continuously with meaningful messages.** The challenge requires that the repository be created and completed inside the contest window. A commit history that shows the weekend of work is itself evidence of compliance.

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NAMING

The project is GLASSPOCKET, one word, all caps in headings, lowercase `glasspocket` in code identifiers, database name `GLASSPOCKET`. The name refers to the "glass pockets" principle in philanthropy, the idea that an organisation handling public generosity should be transparent enough that anyone can see into its pockets.

01 / Mission and judging context

What is being built, why, and against which scoring rubric.

The problem, in one paragraph

The most common charitable fraud is not an invented organisation. It is a real organisation with the serial numbers filed off: a near-identical name, a copied mission statement, a cloned campaign page with one extra character in the URL. String matching cannot catch this because the strings are deliberately different. Meanwhile, on the disbursement side, aid that is genuinely raised is frequently diverted before it reaches a beneficiary, and the sector's answer is a PDF annual report published nine months later. GLASSPOCKET attacks both halves: it detects impersonation by meaning rather than spelling, it detects diversion by geography and timing, it publishes the resulting statistics under a mathematical privacy guarantee so that beneficiaries cannot be re-identified, and it writes a public, tamper-evident receipt for every verified disbursement so a donor can check the claim without trusting anybody.

Evidence base to cite in the application and the write-up

These figures must appear in the application's Brief tab and in the DEV.to post. Attribute each precisely. Do not round, do not embellish, do not add figures that are not on this list without verifying them first.

CLAIM	FIGURE	SOURCE
Charitable and crowdfunding fraud, United States, 2024	More than 4,500 complaints reporting approximately 96 million USD in losses to fraudulent charities, crowdfunding accounts and disaster relief campaigns	FBI IC3 Public Service Announcement I-011625, 16 Jan 2025. ic3.gov/PSA/2025/PSA250116
Impersonation surges after disasters	119 domains registered between 8 and 13 January 2025 using keywords including "LA fire", "wildfire", "relief", "fund" and "rebuild"	BforeAI threat research report on the Los Angeles wildfires
Scale of the legitimate pool being imitated	More than 250 million USD raised for Los Angeles wildfire relief	GoFundMe public statement
Last-mile diversion, Gaza	590 trucks moved from Ashdod to Kerem Shalom, of which 371 were collected inside Gaza; organised criminal looting estimated by field monitors at about 20 percent of cases	WFP State of Palestine External Situation Report 55, 6 June 2025
Warehouse-scale diversion, Sudan	Looting of a Gezira State warehouse holding over 2,500 metric tons of food, enough for nearly 1.5 million people for one month	WFP statement, December 2023
Programme leakage, India	Total financial misappropriation of 169.75 crore rupees with only 20.93 crore recovered, a recovery rate of 12.33 percent, across 125,602 reported cases	MGNREGA action report, as of 29 March 2025
Diversion before delivery, India	86.20 lakh rupees intended for 159 genuine beneficiaries diverted to other bank accounts	CAG audit of PMAY-Gramin, Uttar Pradesh, tabled December 2025

SOURCING DISCIPLINE

The IC3, WFP, WHO, MGNREGA and CAG figures are primary or official. Present them as such. If any additional statistic is introduced during the build, it must carry a named source in the same table, or it must be cut. A project about accountability that cites loosely has already lost the argument.

Scoring rubric this build is optimised against

CRITERION	HOW THIS BUILD WINS IT
Relevance to theme	The challenge theme is generosity, tied to the International Day of Charity. The suggested angles include ethical and accountable giving, and equity and inclusion. GLASSPOCKET sits directly on the first and reaches the second through the privacy layer.
Creativity	The application is structured as four adversarial roles rather than a dashboard. The viewer is invited to be fooled, then to audit, then to attack the system, then to verify it without trusting it.
Technical execution	Differential privacy policies, H3 geospatial indexing, on-warehouse vector similarity, native semantic views and Time Travel, combined with a working on-chain receipt ledger.
Writing quality	Section 12 supplies the narrative spine, the hook and the honesty section.
Use of prize technology	Snowflake is load-bearing: remove it and the project cannot exist. Solana is load-bearing for the one property a warehouse cannot provide, which is public verifiability by a party who does not trust the warehouse operator.

Explicitly out of scope

- **Google AI and Gemini.** Removed from the plan. Do not add an LLM call to the critical path. The Snowflake story is stronger undiluted.
- **User accounts, authentication, payments.** This is an analytical application, not a donation platform. Do not build a checkout.
- **Mainnet.** Devnet only, and say so plainly in the interface and the write-up.
- **Any claim that a specific real organisation is fraudulent.** Real organisations appear only as legitimate reference entities. Every flagged entity is synthetic. This is a legal and ethical hard line, see Section 05.

02 / Hard constraints and known traps

Verified platform limits. Each of these has cost a builder hours. Read before provisioning.

TRAP 01 / UNRECOVERABLE

Select Enterprise Edition at Snowflake signup. Differential privacy requires Enterprise Edition or higher. Cloud platform, region and edition are chosen during signup and cannot be changed afterwards. Choosing Standard means the centrepiece feature of this build is permanently unavailable on that account and the only remedy is creating a new trial. Verify the edition immediately after login with `SELECT CURRENT_EDITION();` before writing any other code.

TRAP 02 / ARCHITECTURAL

Trial accounts have no external network access. Any attempt to create an external access integration or call out from a UDF fails with error 509009. This is why the Solana bridge is an outbound-polling Node process that reaches into Snowflake, rather than Snowflake reaching out to an RPC. Do not attempt to invert this.

TRAP 03 / SILENT

Streamlit in Snowflake enforces a Content Security Policy that blocks scripts, styles, fonts and iframes from any external domain. A Geist font linked from a CDN will silently fall back to a system font. The font must be embedded as base64 inside the stylesheet, see Section 03. Third-party components that make external calls, including pyvis and streamlit-agraph, will not work. Use Plotly and pydeck, which are available in the Snowflake Anaconda channel.

TRAP 04

Map tiles require accepting External Offerings Terms. `st.map` and `st.pydeck_chart` draw tiles from Carto on current Streamlit versions, and Carto is a third-party offering. Accept the terms in Snowsight in the first ten minutes of the build, otherwise the geospatial tab renders empty with no obvious error.

TRAP 05

Differential privacy is incompatible with several things you will want to do. A privacy policy cannot coexist with an aggregation policy or a masking policy on the same object. A privacy-protected table blocks row-level SELECT entirely. A Dynamic Table that depends on a privacy-protected object will fail to refresh. Consequence: the privacy policy attaches only to a terminal serving view at the very end of the pipeline, never to a table in the middle of the DAG.

TRAP 06

Vector similarity at general availability performs an exact scan. A self-join across an organisation table with no pre-filter is a full cross product and will consume the credit budget in a single query. Every similarity query in this specification carries a metadata pre-filter. Do not remove it.

TRAP 07

Streamlit in Snowflake has a 32 MB limit on data passed from backend to frontend in a single command. Never render an unbounded dataframe. Every table in this specification carries an explicit LIMIT.

TRAP 08

Compressed NFT reads require a DAS-enabled RPC endpoint. The default public devnet endpoint does not serve the Digital Asset Standard API. Use a Helius devnet endpoint. Parse asset IDs only from transactions at finalized commitment, and poll with retries because there is an indexing delay between mint and DAS availability. Mitigation: mint the demo receipts hours before recording, never live.

TRAP 09

The trial credit budget is finite. Set warehouse auto-suspend to 60 seconds immediately after account creation. Use an XSMALL warehouse for everything in this build. Embeddings are generated once, in a single batch, and never regenerated on application load. If an embedding query is written inside a Streamlit render path, that is a defect.

Environment matrix

COMPONENT	SETTING	REASON
Snowflake edition	Enterprise	Differential privacy gate
Cloud and region	AWS, region nearest the operator	Latency during a live demo
Warehouse	XSMALL, auto-suspend 60s, auto-resume true	Credit conservation
Database	GLASSPOCKET	Naming contract
Schemas	RAW, STAGING, MARTS, SERVING, ORACLE	Privacy policy isolation
Streamlit runtime	Warehouse runtime, Python 3.11	Simplest path, Plotly and pydeck available
Solana cluster	Devnet	Free, and honest to disclose
Node version	20 LTS or higher	Umi and Bubblegum requirement

03 / Design system

The application must not look like a default Streamlit app. This section is not decoration, it is a scored criterion.

Typography

The single typeface is **Geist**, the Vercel sans-serif. It is used for every element without exception: headings, body, labels, numerals, table cells, chart axis labels, chart tooltips and code-like strings such as asset identifiers and SQL fragments.

ABSOLUTE RULE / NO MONOSPACE

No monospaced typeface appears anywhere in the application. Not Geist Mono, not Source Code Pro, not the browser default. Where the interface displays a Solana asset identifier, a wallet address, an EIN or a SQL snippet, it is set in Geist with the OpenType tabular-figures feature enabled and a tinted background block. Any element that resolves to a monospace font is a defect. Explicitly override Streamlit's built-in code styling; do not rely on `st.code`, which forces monospace. Build a custom rendering helper instead, see below.

Embedding Geist under the Content Security Policy

Because external font hosts are blocked, Geist must be inlined. Obtain the woff2 files from the npm package `geist`, subpath `dist/fonts/geist-sans/`. Required weights: 400 Regular, 500 Medium, 600 SemiBold, 700 Bold. Base64 each file and emit a single stylesheet at application start.

```
# tools/build_font_css.py, run once, output committed to the repo
import base64, pathlib
WEIGHTS = {"Regular": 400, "Medium": 500, "SemiBold": 600, "Bold": 700}
faces = []
for name, weight in WEIGHTS.items():
    raw = pathlib.Path(f"fonts/Geist-{name}.woff2").read_bytes()
    b64 = base64.b64encode(raw).decode()
    faces.append(
        "@font-face{font-family:'Geist';font-style:normal;"
        f"font-weight:{weight};font-display:block;"
        f"src:url(data:font/woff2;base64,{b64}) format('woff2');}"
    )
pathlib.Path("app/assets/geist.css").write_text("".join(faces))
```

Inject it in the application, together with the global override, as the very first Streamlit call after page configuration:

```
# app/theme.py
import streamlit as st, pathlib

def inject_theme():
    font_css = pathlib.Path("assets/geist.css").read_text()
    st.markdown(f"<style>{font_css}{GLOBAL_CSS}</style>", unsafe_allow_html=True)

GLOBAL_CSS = """
html, body, [class*="css"], [data-testid], button, input, textarea, select,
code, pre, kbd, samp, .stMarkdown, .stDataFrame, .js-plotly-plot, .plotly {
  font-family: 'Geist', -apple-system, sans-serif !important;
  font-feature-settings: 'tnum' 1, 'cv01' 1;
}
```

```
h1,h2,h3,h4 { letter-spacing: -0.028em; font-weight: 650; }
#MainMenu, footer, header { visibility: hidden; }
.block-container { padding-top: 2.4rem; max-width: 1180px; }
""""
```

Type scale

TOKEN	SIZE	WEIGHT	TRACKING	USE
display	44 px	800	-0.035em	Tab hero numbers, the single big figure on each tab
title	28 px	700	-0.03em	Tab titles
heading	18 px	650	-0.02em	Section headings inside a tab
body	14 px	400	-0.005em	Paragraphs and table cells
label	11 px	600	0.09em, uppercase	Metric captions, chart eyebrows
data	13 px	500	0	Identifiers, hashes, SQL, with tabular figures

Rendering mode

LIGHT MODE ONLY

The application renders in light mode exclusively. White surfaces, ink text, colourful data. Do not build a dark theme, do not offer a theme toggle, and do not rely on the viewer's Snowsight preference. Set the Streamlit theme explicitly so a viewer with dark mode enabled still sees the intended design. Charts carry the colour; the page does not. This is a deliberate inversion of the usual data-application look and it is what will make the screenshots stand out in a feed of dark dashboards.

In `.streamlit/config.toml` : base is light, backgroundColor is #FFFFFF, secondaryBackgroundColor is #F7F8FA, textColor is #16181D, primaryColor is #29B5E8, font is sans serif.

Colour



Ink

#08090B. Page background in dark surfaces, primary text on light.



Snowflake Blue

#29B5E8. Primary accent, verified states, active tab.

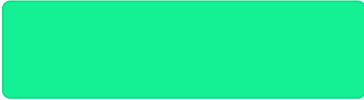


Flag Red

#E5484D. Detected impersonation, impossible delivery, missing receipt.



Solana Purple
#9945FF. On-chain elements only. Never used for anything else.



Solana Green
#14F195. Confirmed on-chain receipt.



Noise Amber
#F5A623. Differential privacy noise, budget consumption.

TOKEN	HEX	APPLICATION
surface	#FFFFFF	Card and page background
surface-sunken	#F7F8FA	Data blocks, identifier chips
border	#E4E7EB	All 1 px rules and card edges
text-muted	#6B7280	Captions, secondary labels
truth	#16181D	The true value in privacy comparisons
noise	#F5A623	The differentially private value

Categorical palette for infographics

Charts with more than two series draw from this ordered palette. Use it in order, never shuffle, so the same category keeps the same colour across all eleven tabs. Every value is checked for legibility on white.

ORDER	HEX	NAME	TYPICAL ASSIGNMENT
1	#29B5E8	Snowflake Blue	Verified, plausible, primary series
2	#9945FF	Solana Purple	On-chain, receipts
3	#F5A623	Noise Amber	Privacy noise, uncertainty, seeded data
4	#E5484D	Flag Red	Needs a second look, unaccounted
5	#17A34A	Confirmed Green	Delivered, receipt found
6	#0F7B8A	Deep Teal	Fourth and later categorical series
7	#B45BD6	Orchid	Fifth categorical series
8	#6B7280	Neutral	Baseline, context, everything unremarkable

Infographic components to build once and reuse

COMPONENT	DEFINITION
hero(value, caption)	Oversized figure in the display token with a label-token caption beneath. Opens every tab without exception.

stat_band(items)	Row of three to five bordered cards, each a figure plus caption. No shadows, 1 px borders, equal widths.
identifier(text)	Renders a hash, EIN or asset ID in Geist with tabular figures on a sunken background with a copy control. This is the replacement for st.code and it is why no monospace appears.
chip(text, tone)	Pill in one of five tones: verified, seeded, flagged, chain, neutral.
step_explainer(steps)	Three or four numbered blocks in a horizontal row, each with a short title and one sentence. Used at the foot of most tabs to explain the mechanism in plain language.
compare_columns(left, right)	Two-column layout with a divider, used for true against private on Tab 05 and before against after on Tab 07.
source_note(text, url)	Small muted line with a link, placed under any figure that came from a cited source. Every cited number in the interface carries one.

Layout primitives

- **Spacing scale.** 4, 8, 12, 16, 24, 32, 48, 64 px. No arbitrary values.
- **Radius.** 6 px on cards, 4 px on chips and inputs, 999 px on pills. Charts have no radius.
- **Elevation.** None. Separation comes from a single 1 px border in the border token. No drop shadows anywhere.
- **Grid.** Twelve columns implemented with `st.columns`. Maximum content width 1180 px.
- **Hero pattern.** Every tab opens with one oversized figure in the display token, one line of context beneath it in the label token, and nothing else above the fold. This is what converts a dashboard into a narrative.

Chart styling contract

Every Plotly figure passes through a single helper so that styling cannot drift. Implement it once in `app/charts.py` and route all figures through it.

```
def style_fig(fig, height=340, legend=False):
    fig.update_layout(
        font=dict(family="Geist", size=13, color="#16181D"),
        paper_bgcolor="rgba(0,0,0,0)", plot_bgcolor="rgba(0,0,0,0)",
        margin=dict(l=8, r=8, t=28, b=8), height=height,
        showlegend=legend, hoverlabel=dict(font_family="Geist", font_size=13),
        xaxis=dict(showgrid=False, zeroline=False, linecolor="#E4E7EB"),
        yaxis=dict(gridcolor="#F0F2F4", zeroline=False, linecolor="#E4E7EB"),
    )
    return fig
```

CHART RULES

No chart titles inside the figure; the title is a Streamlit heading above it, so typography stays consistent. No gridlines on the x axis. No legend unless two or more series genuinely overlap. Colour carries meaning strictly according to the palette table, so a red bar always means flagged and an amber line always means privacy noise. Animation is used exactly twice in the whole application, both specified in Section 07.

04 / Repository layout

Exact file tree. Create it in full before writing feature code.

```
glasspocket/
  README.md          # challenge disclosure, setup, honesty section
  .gitignore
  .env.example       # never commit a real key
  sql/
    00_account_setup.sql    # warehouse, database, schemas, edition assert
    01_raw_tables.sql
    02_load_bmf.sql         # IRS Business Master File
    03_load_iati.sql        # IATI activity and location data
    04_seed_adversaries.sql  # synthetic clones, labelled
    05_staging_dynamic_tables.sql
    06_embeddings.sql       # AI_EMBED batch, run once
    07_clone_detection.sql   # vector similarity
    08_geospatial_h3.sql
    09_risk_score_udf.sql
    10_semantic_view.sql
    11_privacy_policy.sql    # differential privacy, run first in practice
    12_serving_views.sql
    13_oracle_queue.sql
    99_acceptance_checks.sql
  bridge/
    package.json
    src/index.ts           # poll queue, mint, write back
    src/snowflake.ts
    src/tree.ts            # one-time Merkle tree creation
    src/mint.ts
  app/
    glasspocket_app.py     # entrypoint, tab router
    theme.py
    charts.py
    queries.py             # every SQL string, one place
    components.py          # hero, metric, identifier chip, comparison card
    assets/geist.css       # generated, committed
  tabs/
    tab_00_brief.py
    tab_01_donor.py
    tab_02_auditor.py
    tab_03_attacker.py
    tab_04_skeptic.py
    tab_05_historian.py
  tools/
    build_font_css.py
    make_synthetic.py
  docs/
    demo_script.md
    architecture.txt
```

CONVENTION

Every SQL statement the application executes lives as a named constant in `app/queries.py`. No inline SQL inside a tab module. This makes it possible to run the entire query surface headlessly during acceptance testing, and it makes the write-up easy because the SQL is quotable from one file.

04B / Environment and secrets

Every variable the build needs, where it comes from, and which process reads it.

RULE

No secret is ever committed, printed to a log, pasted into a Streamlit page, or visible in the demo recording. Commit `.env.example` with empty values only. Add `.env` and `*.json` keypair files to `.gitignore` before the first commit, not after. If a key is ever exposed, rotate it and note the rotation in the README rather than quietly rewriting history.

Bridge process variables

VARIABLE	EXAMPLE SHAPE	WHERE TO OBTAIN IT
SNOWFLAKE_ACCOUNT	abcdefg-xy12345	Snowsight, account details panel. Format is orgname-accountname. This is the single most commonly mistyped value in the whole build.
SNOWFLAKE_USER	the signup username	Chosen at account creation
SNOWFLAKE_PASSWORD	secret	Chosen at account creation. Prefer key-pair authentication if multi-factor is enforced on the trial, which it may be.
SNOWFLAKE_PRIVATE_KEY_PATH	./keys/rsa_key.p8	Optional alternative to password. Generate with openssl, register the public key with ALTER USER SET RSA_PUBLIC_KEY.
SNOWFLAKE_ROLE	ACCOUNTADMIN	The bridge writes to ORACLE schema and needs write privileges. Do not reuse GP_ANALYST here.
SNOWFLAKE_WAREHOUSE	GP_WH	Created in 00_account_setup.sql
SNOWFLAKE_DATABASE	GLASSPOCKET	Fixed by the naming contract
SNOWFLAKE_SCHEMA	ORACLE	Queue and mint log live here
SOLANA_RPC_URL	a Helius devnet endpoint including the API key	Helius dashboard, free tier. Must be DAS-enabled. The default public devnet endpoint will not work for reads.
SOLANA_KEYPAIR_PATH	./keys/authority.json	Generate locally with the Solana CLI. This wallet is the tree authority and pays for mints. Fund from a devnet faucet on Friday, not Saturday.
MERKLE_TREE_ADDRESS	base58 public key	Written once by tree.ts on first run, then pasted back into the environment. Tree parameters are immutable, so record this carefully.
COLLECTION_MINT	base58 public key	

Created once alongside the tree so receipts group under a single collection in wallets and explorers

RECEIPT_SALT	a long random string	Generate locally. Used to hash organisation identifiers before they go on chain, so the ledger carries a claim rather than a directly reversible identifier.
MINT_BATCH_SIZE	50	Tuning value, not a secret
MINT_POLL_MS	4000	Tuning value, not a secret

Application variables

The Streamlit application runs inside Snowflake and therefore needs no credentials at all. It obtains its session with `get_active_session()` from `snowflake.snowpark.context`. If any Snowflake credential appears in the application directory, that is a defect. The only application-side configuration values are:

VALUE	DEFAULT	PURPOSE
GP_ANALYST_ROLE	GP_ANALYST	The role the Attacker tab switches to so the privacy policy actually applies. Set with a session USE ROLE at the top of that tab and restore afterwards.
SIMILARITY_THRESHOLD	0.86	Production cosine cut-off, displayed in the interface rather than hidden
EXPLORER_BASE	explorer.solana.com with the devnet cluster parameter	Link construction only. No network call at render time.
DEMO_MODE	true	When true, the Attacker tab exposes the budget reset control and the Historian tab exposes the write control. State in the interface that a production deployment would expose neither.

Local tooling prerequisites

TOOL	VERSION	USED FOR
Node	20 LTS or later	Bridge, Umi, Bubblegum
Solana CLI	current stable	Keypair generation, devnet airdrop, balance checks
Python	3.11	Synthetic data generation, PCA projection, graph layout, font encoding. Matches the Streamlit warehouse runtime.
Python packages, local only	pandas, numpy, scikit-learn, networkx	Offline precomputation. None of these need to exist inside Snowflake, because their outputs are written to tables.
SnowSQL or the Snowsight worksheet	current	Running the numbered SQL files in order

04C / Data provenance map

Which tab draws from which source. Use this to sequence the loads so no tab is blocked waiting on data it does not need.

TAB	NAME	READS FROM	ULTIMATE SOURCE
00	The Brief	MARTS counts, V_PIPELINE_HEALTH, a static cited-evidence table	All sources, plus the seven citations hand-entered from Section 01
01	Give With Confidence	CLONE_CONFIRMED, ORG_PROJECTION, ORG_RISK	IRS Business Master File for the real side, synthetic generator for the suspect side
02	The Trust Graph	GRAPH_NODES, GRAPH_EDGES, THRESHOLD_CURVE, CLONE_PAIRS	Same as Tab 01, plus offline networkx layout
03	Follow The Money	V_HEX_RISK, DELIVERY_GEOMETRY	IATI Datastore activity locations, plus synthetic disbursements
04	The Last Mile	V_ATTRITION_FLOW, DELIVERY_GEOMETRY, DISTRICT_ATTRITION	IATI transactions, synthetic delivery events calibrated to the WFP figures
05	The Wall	SERVING.V_BENEFICIARY_OUTCOMES protected, and its unprotected twin	Entirely synthetic beneficiary records. No real personal data enters this project at any point.
06	The Receipt	ORACLE.MINT_LOG, MINT_QUEUE, V_RECEIPT_GAP	Solana devnet, mirrored locally by the bridge
07	The Historian	MARTS.ORG_RISK at successive Time Travel offsets	Snowflake internal retention
08	Ask The Warehouse	MARTS.GIVING_SEMANTICS	Semantic layer over staging and marts
09	Where A Dollar Lands	V_CONFIDENCE_RANK	IRS Business Master File only. No synthetic entity appears on this tab.
10	Method And Honesty	Batch identifier counts across every table	Computed live so the honesty table cannot drift from reality

Acquisition order and effort

SOURCE	EFFORT	BLOCKS TABS	NOTES FOR THE IMPLEMENTING AGENT
IRS EO Business Master File	20 minutes	01, 02, 09	Regional CSV files from the IRS Charities and Non-Profits section. Download all regions, concatenate, load with a file format that tolerates ragged rows. Column names differ slightly between regions, so normalise on load rather than trusting a single header.

IRS Auto-Revocation List	10 minutes	01, 09	Same IRS section. Join on EIN. Describe as loss of good standing, never as wrongdoing.
IATI Datastore	45 minutes	03, 04	Take a bounded slice, for example one sector or two countries, rather than the full corpus. The full datastore is far larger than this build needs and loading it will waste an hour you do not have.
Snowflake Marketplace free listing	5 minutes	none, enrichment only	Attach at least one. It is a linked resource on the challenge page, which means judges are looking for it. A geography or demographics listing joins cleanly to the district dimension.
ProPublica Nonprofit Explorer	20 minutes	09 detail panel	Fetch a few hundred records during preparation and load as a table. Never call at render time.
Synthetic generator	60 minutes	01, 02, 04, 05	The largest single piece of preparation work. Budget properly for it, because the quality of the adversaries determines whether the detection looks impressive or trivial.

IF A SOURCE WILL NOT LOAD

The Business Master File is the only genuinely load-bearing external dataset. If IATI proves awkward inside the time budget, generate the disbursement layer entirely synthetically, calibrate it against the published WFP figures, and disclose that on Tab 10. Tabs 03 and 04 remain fully functional. Do not spend more than ninety minutes fighting an upstream data format on a weekend deadline.

05 / Data layer and sources

Where every row comes from, how it is loaded, and the labelling contract that keeps this project honest.

Source inventory

SOURCE	WHAT IT GIVES	ACCESS	LOAD PATH
IRS Exempt Organizations Business Master File	Every registered tax-exempt organisation in the United States. EIN, legal name, street, city, state, ZIP, NTEE classification code, revenue and asset amounts. Roughly 1.8 million rows across regional files.	Public CSV, no authentication. irs.gov, Charities and Non-Profits, EO BMF extract	Download, upload through Snowsight, COPY INTO. Fastest path, no XML parsing. This is the primary corpus.
IRS Auto-Revocation List	Organisations whose exempt status was revoked. Used as a real negative label.	Public CSV from the same IRS page	COPY INTO a small dimension table
IATI Datastore	Development and humanitarian activities with budgets, transactions, sectors and geocoded activity locations. This is the source of the geospatial layer.	Public API and bulk download. iatistandard.org, IATI Datastore	Pull a country or sector slice as CSV, COPY INTO
ProPublica Nonprofit Explorer	Per-organisation filing detail for the drill-down panel.	Public JSON API	Fetch a small set during preparation, load as a table. Not called at runtime.
Snowflake Marketplace, free listings	At least one free listing attached to the account, for geography or demographic enrichment.	Snowsight Marketplace, filter to free	Attach as a shared database, join directly. Zero load time.

BE PRECISE ABOUT THE REVOCATION LIST

The Auto-Revocation List is overwhelmingly a record of organisations that failed to file for three consecutive years. It is not a fraud list. The application must describe it as "no longer in good standing" and never as "fraudulent". Mislabelling it would be exactly the kind of sloppy inference this project exists to criticise.

The synthetic adversary set, and the labelling contract

There is no public labelled dataset of fake charities, because fakes are taken down rather than catalogued. Therefore the impersonating organisations in this build are generated. This is acceptable and defensible only if it is disclosed relentlessly.

Generation rules

`tools/make_synthetic.py` produces roughly 400 adversarial organisations by sampling real BMF entries and applying transformations that mirror documented real-world techniques:

- **Semantic clone.** Replace one word with a synonym while preserving meaning, for example Relief becomes Aid, Foundation becomes Trust, Children becomes Kids.
- **Token reorder.** Same words, different order, plus a filler word such as National or United.
- **Homoglyph and spacing.** A doubled letter, a hyphen inserted, an abbreviation expanded.
- **Disaster attachment.** Append a current disaster keyword to a legitimate base name, mirroring the documented post-wildfire domain pattern.
- **Geographic implausibility.** Assign disbursement coordinates far outside any plausible operating radius for the organisation's registered address.

The contract

RULE	IMPLEMENTATION
Every synthetic row is machine-identifiable	Column <code>batch_id</code> equals <code>'SYNTH_ADVERSARY_V1'</code> . Real rows carry <code>'IRS_BMF_2026'</code> or <code>'IATI_2026'</code> .
Every synthetic row is human-identifiable	Column <code>is_synthetic</code> boolean, surfaced in the interface as a small amber chip reading "seeded".
No real organisation is ever flagged	The clone detection query hard-filters so that the suspect side of a pair must satisfy <code>is_synthetic = TRUE</code> . Real organisations appear only as the legitimate target of an impersonation.
The interface states it	A persistent banner in the application header and a dedicated panel on the Brief tab.
The write-up states it	A section headed "What is real and what is simplified", with a two-column table.

NON-NEGOTIABLE

Under no circumstances may the application display a real, named organisation in a flagged, suspicious or fraudulent state. The suspect side of every detected pair is synthetic by construction, enforced in SQL, not merely by convention. Add this as an acceptance check that fails the build if violated.

Core schema

```
-- sql/O1_raw_tables.sql, abbreviated to the columns the app actually uses
```

```
CREATE OR REPLACE TABLE STAGING.ORGs (  
  org_id    STRING    PRIMARY KEY,  
  ein       STRING,  
  name      STRING,  
  blurb     STRING,  
  city      STRING,  
  state     STRING,  
  ntee_code STRING,  
  cause     STRING,  
  lat       FLOAT,  
  lon       FLOAT,
```

```

home_h3    STRING,    -- resolution 5 cell of registered address
region_h3  STRING,    -- resolution 3 cell, used as the similarity pre-filter
max_hops   INT,       -- plausible operating radius in grid steps
is_verified BOOLEAN,   -- present in BMF and in good standing
is_synthetic BOOLEAN,
batch_id   STRING,
first_seen TIMESTAMP_NTZ,
name_vec   VECTOR(FLOAT, 768)
);

CREATE OR REPLACE TABLE STAGING.DISBURSEMENTS (
  disbursement_id STRING PRIMARY KEY,
  org_id           STRING,
  programme_code   STRING,
  beneficiary_id   STRING, -- entity key for differential privacy
  amount_usd       NUMBER(12,2),
  pledged_usd      NUMBER(12,2),
  district         STRING,
  lat              FLOAT,
  lon              FLOAT,
  delivery_h3      STRING, -- resolution 7
  dispatched_at    TIMESTAMP_NTZ,
  delivered_at     TIMESTAMP_NTZ,
  status           STRING, -- PLEDGED, DISPATCHED, DELIVERED, UNACCOUNTED
  is_synthetic     BOOLEAN,
  batch_id         STRING
);

```

06 / Snowflake feature stack

Six features carry this build. Presented in headline order, which is also the order they appear in the write-up.

06.1 Differential privacy, the centrepiece

This is the feature that distinguishes the submission. It is rare in hackathon work, it is Enterprise-gated, and it converts the ethics discussion from a disclaimer into the technical thesis. It is also the riskiest item, which is why it is built first.

```
-- sql/t1_privacy_policy.sql
USE ROLE ACCOUNTADMIN;

-- The analyst persona the application connects as
CREATE ROLE IF NOT EXISTS GP_ANALYST;

CREATE OR REPLACE PRIVACY POLICY GLASSPOCKET.SERVING.beneficiary_policy
AS () RETURNS privacy_budget ->
CASE
  WHEN CURRENT_ROLE() = 'ACCOUNTADMIN' THEN no_privacy_policy()
  WHEN CURRENT_ROLE() = 'GP_ANALYST' THEN privacy_budget(name => 'public_analyst')
  ELSE privacy_budget(name => 'default')
END;

-- Attach ONLY to the terminal serving view, never mid-pipeline
CREATE OR REPLACE VIEW GLASSPOCKET.SERVING.V_BENEFICIARY_OUTCOMES AS
SELECT beneficiary_id, district, programme_code, cause,
       amount_usd, delivered_flag, month_key
FROM GLASSPOCKET.MARTS.BENEFICIARY_FACTS;

ALTER VIEW GLASSPOCKET.SERVING.V_BENEFICIARY_OUTCOMES
SET PRIVACY POLICY GLASSPOCKET.SERVING.beneficiary_policy
ENTITY KEY (beneficiary_id);

-- Privacy domains keep the noise calibrated and prevent range inference
ALTER VIEW GLASSPOCKET.SERVING.V_BENEFICIARY_OUTCOMES
MODIFY COLUMN amount_usd SET PRIVACY DOMAIN (0, 5000);
ALTER VIEW GLASSPOCKET.SERVING.V_BENEFICIARY_OUTCOMES
MODIFY COLUMN district SET PRIVACY DOMAIN (SELECT DISTINCT district FROM MARTS.DISTRICT_DIM);

GRANT SELECT ON VIEW GLASSPOCKET.SERVING.V_BENEFICIARY_OUTCOMES TO ROLE GP_ANALYST;
```

The paired unprotected view. The Attacker tab shows true values beside private values, so a second view over the same facts is required, readable only by a privileged role and used exclusively to render the left-hand comparison column. Name it `SERVING.V_BENEFICIARY_OUTCOMES_TRUE` and label it unmistakably in the interface as the counterfactual that would exist without the policy.

VERIFICATION STEP, DO NOT SKIP

Immediately after applying the policy, connect as GP_ANALYST and run a row-level select. It must fail. Then run a broad aggregate; it must succeed with slight noise. Then run an aggregate filtered to a single beneficiary; the noise must be large or the query must be refused. If all three behave as described, the centrepiece works and the rest of the build is downhill. If not, apply the fallback in Section 10 within thirty minutes and move on.

06.2 On-warehouse embeddings and vector similarity

```
-- sql/06_embeddings.sql, run ONCE, never inside the app
UPDATE STAGING.ORGs
  SET name_vec = AI_EMBED('snowflake-arctic-embed-m',
    name || ' :: ' || COALESCE(blurb, '') || ' :: ' || COALESCE(cause, ''))
WHERE name_vec IS NULL;

-- sql/07_clone_detection.sql
CREATE OR REPLACE TABLE MARTS.CLONE_PAIRS AS
SELECT
  s.org_id AS suspect_id, s.name AS suspect_name,
  t.org_id AS target_id, t.name AS target_name,
  s.city, s.state, s.cause,
  VECTOR_COSINE_SIMILARITY(s.name_vec, t.name_vec) AS semantic_sim,
  JAROWINKLER_SIMILARITY(s.name, t.name) / 100.0 AS string_sim,
  -- the gap between meaning and spelling is the whole thesis
  VECTOR_COSINE_SIMILARITY(s.name_vec, t.name_vec)
  - (JAROWINKLER_SIMILARITY(s.name, t.name) / 100.0) AS evasion_gap
FROM STAGING.ORGs s
JOIN STAGING.ORGs t
  ON s.region_h3 = t.region_h3 -- REQUIRED pre-filter, see Trap 06
 AND s.cause = t.cause
 AND s.ein <> t.ein
WHERE s.is_synthetic = TRUE -- suspect side is ALWAYS seeded
 AND t.is_verified = TRUE -- target side is ALWAYS a real, good-standing org
QUALIFY VECTOR_COSINE_SIMILARITY(s.name_vec, t.name_vec) >= 0.86;
```

Why evasion_gap matters. A high semantic similarity paired with a low string similarity is the signature of deliberate evasion: the impersonator kept the meaning and changed the spelling on purpose. This single derived column is the most quotable number in the entire project and it drives the primary chart on the Donor tab.

06.3 Semantic confirmation with AI_FILTER

Applied only to the already-narrowed candidate set, so cost stays trivial. This demonstrates an AI predicate used as a join condition, which plain SQL cannot express.

```
CREATE OR REPLACE TABLE MARTS.CLONE_CONFIRMED AS
SELECT cp.*, TRUE AS ai_confirmed
FROM MARTS.CLONE_PAIRS cp
WHERE AI_FILTER(
  'These two nonprofit names describe organisations a reasonable donor could confuse ' ||
  'for one another. First: ' || cp.suspect_name || '. Second: ' || cp.target_name);
```

06.4 H3 geospatial diversion detection

```
-- sql/08_geospatial_h3.sql
CREATE OR REPLACE TABLE MARTS.DELIVERY_GEOMETRY AS
SELECT
  d.disbursement_id, d.org_id, d.district, d.amount_usd, d.status,
  d.delivery_h3,
  H3_GRID_DISTANCE(o.home_h3, H3_CELL_TO_PARENT(d.delivery_h3, 5)) AS hops,
  o.max_hops,
  ST_DISTANCE(ST_POINT(o.lon, o.lat), ST_POINT(d.lon, d.lat)) / 1000 AS km_from_base,
```

```

DATEDIFF('hour', d.dispatched_at, d.delivered_at) AS transit_hours,
CASE
  WHEN d.status = 'UNACCOUNTED' THEN 'NEVER_ARRIVED'
  WHEN H3_GRID_DISTANCE(o.home_h3, H3_CELL_TO_PARENT(d.delivery_h3, 5)) > o.max_hops
    THEN 'OUTSIDE_FOOTPRINT'
  WHEN DATEDIFF('hour', d.dispatched_at, d.delivered_at) < 1
    AND ST_DISTANCE(ST_POINT(o.lon,o.lat), ST_POINT(d.lon,d.lat)) > 400000
    THEN 'IMPOSSIBLE_TRANSIT'
  ELSE 'PLAUSIBLE'
END AS geometry_verdict
FROM STAGING.DISBURSEMENTS d
JOIN STAGING.ORGs o USING (org_id);

-- Hex aggregation that feeds the pydeck layer directly
CREATE OR REPLACE VIEW SERVING.V_HEX_RISK AS
SELECT delivery_h3 AS h3,
  COUNT(*) AS n,
  SUM(amount_usd) AS usd,
  AVG(CASE WHEN geometry_verdict <> 'PLAUSIBLE' THEN 1 ELSE 0 END) AS flag_rate
FROM MARTS.DELIVERY_GEOMETRY
GROUP BY 1;

```

06.5 The risk score, decomposable by design

An opaque score is unusable in an accountability tool. The score must be reconstructable by hand from its components, and the interface must show the components. Implement as a SQL UDF so it is inspectable.

```

CREATE OR REPLACE FUNCTION MARTS.F_RISK(
  semantic_sim FLOAT, evasion_gap FLOAT,
  geom_flags INT, unaccounted_ratio FLOAT, missing_receipts INT)
RETURNS FLOAT
AS $$
  LEAST(100,
    35 * GREATEST(0, (semantic_sim - 0.86) / 0.14)
    + 25 * LEAST(1, evasion_gap / 0.45)
    + 20 * LEAST(1, geom_flags / 5.0)
    + 12 * LEAST(1, unaccounted_ratio)
    + 8 * LEAST(1, missing_receipts / 10.0))
$$;

```

06.6 Semantic view and Time Travel

```

-- sql/10_semantic_view.sql
CREATE OR REPLACE SEMANTIC VIEW MARTS.GIVING_SEMANTICS
TABLES (
  orgs AS STAGING.ORGs PRIMARY KEY (org_id)
    WITH SYNONYMS ('charities', 'nonprofits', 'organisations'),
  disb AS MARTS.DELIVERY_GEOMETRY PRIMARY KEY (disbursement_id)
    WITH SYNONYMS ('disbursements', 'deliveries', 'aid shipments')
)
RELATIONSHIPS (disb (org_id) REFERENCES orgs (org_id))
FACTS (disb.amount_usd AS amount_usd)
DIMENSIONS (orgs.state AS state, orgs.cause AS cause,
  disb.geometry_verdict AS verdict)
METRICS (

```

```

disb.total_usd AS SUM(disb.amount_usd)
WITH SYNONYMS ('total disbursed', 'money moved'),
disb.flagged_pct AS AVG(CASE WHEN disb.geometry_verdict <> 'PLAUSIBLE'
THEN 1 ELSE 0 END)
WITH SYNONYMS ('flag rate', 'share flagged')
);

```

Time Travel as tamper evidence. Enterprise Edition allows up to ninety days of Time Travel. The Historian tab uses it to demonstrate that a verdict cannot be quietly rewritten. The mechanic: update a row, then query the same row as it existed before the update, and show both.

```

SELECT 'current' AS version, org_id, risk_score FROM MARTS.ORG_RISK WHERE org_id = ?
UNION ALL
SELECT 'before', org_id, risk_score
FROM MARTS.ORG_RISK AT(OFFSET => -300) WHERE org_id = ?;

```

Feature-to-tab map

SNOWFLAKE FEATURE	WHERE THE JUDGE SEES IT	PRIORITY
Privacy policy, differential privacy	Tab 03, Attacker	must ship
AI_EMBED and VECTOR_COSINE_SIMILARITY	Tab 01, Donor	must ship
H3 grid functions and ST_DISTANCE	Tab 02, Auditor	must ship
SQL UDF, decomposable score	Tabs 01 and 02	must ship
AI_FILTER as a semantic predicate	Tab 01, confirmation chip	high
Dynamic Tables with TARGET_LAG	Tab 00, pipeline strip	high
Semantic view and Cortex Analyst	Tab 00, question box	cut third
Time Travel	Tab 05, Historian	cut second

07 / Application specification

Eleven tabs. Each carries six or seven sections, a skeleton sketch, its own data provenance and its own acceptance list.

The narrative spine

This is not a dashboard. It is an argument in eleven moves, ordered so a viewer clicking straight through experiences a beginning, a turn and a payoff. State the spine at the top of Tab 00 in one sentence: **generosity fails not because people stop giving but because they cannot see where the money went, so this is the machinery that lets them see.**

TAB	NAME	THE MOVE	EMOTIONAL BEAT
00	The Brief	Establish stakes with cited numbers	This is real and it is large
01	Give With Confidence	Telling real from imitation is genuinely hard	Humility. The viewer gets it wrong.
02	The Trust Graph	Imitation is a structure, not incidents	Scale
03	Follow The Money	Geography of where aid actually landed	Orientation
04	The Last Mile	Where the gap between pledged and delivered opens	Loss, quantified
05	The Wall	Invite the viewer to attack the privacy layer	The turn
06	The Receipt	Verify without trusting the operator	Payoff
07	The Historian	The record cannot be quietly rewritten	Durability
08	Ask The Warehouse	Plain-language questions over governed data	Accessibility
09	Where A Dollar Lands	Point the viewer toward giving	Warmth
10	Method And Honesty	Declare what is real and what is seeded	Credibility

FRAMING RULE, EVERY TAB

The product is a confidence layer for people who want to give, not an accusation engine. Interface language reflects this throughout. Write verified, unverified, needs a second look, confidence, receipt found. Never write

fraud, criminal, scam or guilty about any entity. The word fraud appears only on Tab 00 where it quotes a cited source.

Universal tab contract

Every tab, without exception, follows this order. Deviating breaks the rhythm that makes eleven tabs feel like one product.

SLOT	ELEMENT	RULE
1	Hero	One oversized figure and one caption line. Nothing else above the fold. Uses the hero component.
2	Stat band	Three to five supporting figures in bordered cards.
3	Primary visual	The one chart a viewer would screenshot. Full width, minimum 380 px tall.
4	Supporting pair	Two charts side by side that qualify or decompose the primary.
5	Detail surface	A table, inspector panel or drill-down bound to a selection made above.
6	Explainer	Three or four step blocks in plain language describing the mechanism. This is what makes the application legible to a non-technical judge.
7	Provenance footer	Muted line naming the data behind this tab, with a link to Tab 10.

Shell

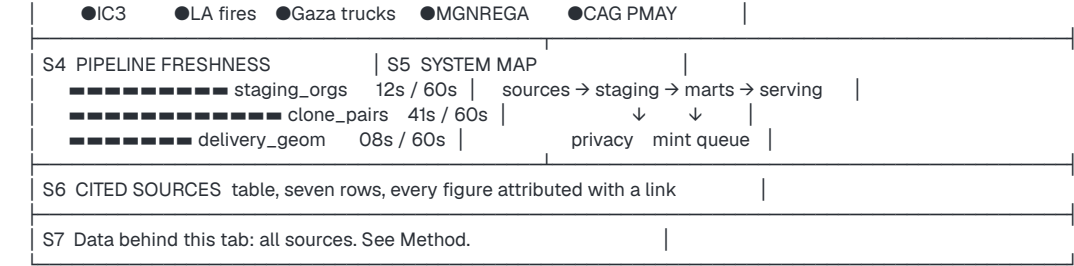
GLASSPOCKET	[seeded data is labelled]	[solana devnet]	
00 Brief	01 Confidence	02 Graph	03 Money
04 Last Mile	05 Wall	06 Receipt	...
[tab content, 1180px max width, always opens with the hero slot]			

TAB 00

The Brief

You want to help. Here is what stands between you and that.

S1 \$96,000,000											reported lost to fraudulent charities, crowdfunding and disaster relief, US, 2024											FBI IC3, January 2025										
S2 orgs			disbursements			receipts on			needs a			pipeline objects																				
indexed			traced			chain			second look			fresh																				
S3 EVIDENCE TIMELINE										2023		—————				2024		—————				2025		—————				2026				



SEC	ELEMENT	COMPONENT	CONTENT
S1	Hero	hero()	The IC3 figure. Largest number in the entire application. Caption carries the source inline.
S2	Stat band	stat_band()	Five live counts from Q00_KPI. The fifth, pipeline objects fresh, quietly advertises the Dynamic Table layer.
S3	Evidence timeline	C00-1	Seven cited incidents plotted over time
S4	Pipeline freshness	C00-2	Refresh age against TARGET_LAG per Dynamic Table
S5	System map	Static SVG infographic	Five-stage flow drawn as inline SVG using palette colours. Not a chart, an illustration. Hand-authored once.
S6	Cited sources	Styled table	The Section 01 evidence table rendered in full with links
S7	Provenance	source_note()	Links to Tab 10

C00-1 / PLOTLY SCATTER WITH ANNOTATIONS

Evidence timeline

X is date from early 2023 to mid 2026. Y is a jittered category lane so markers do not collide. Marker area scales with magnitude, normalised within each unit type since dollars and tonnes are not comparable. Colour by region using the categorical palette in order. Two annotations pinned permanently: the Los Angeles domain surge and the Gaza truck shortfall.

Interaction. Clicking a marker expands a detail card beneath carrying the verbatim figure, the issuing body and the source link. Selection persists so the card stays open while the viewer reads.

C00-2 / PLOTLY HORIZONTAL BAR WITH A REFERENCE LINE

Pipeline freshness

One bar per Dynamic Table. Length is seconds since last refresh, with a dashed vertical rule at the configured TARGET_LAG. Bars inside lag render Snowflake Blue, bars past it render Noise Amber. Reads from INFORMATION_SCHEMA.DYNAMIC_TABLE_REFRESH_HISTORY through the V_PIPELINE_HEALTH view.

Why it earns points. It is how a judge sees the declarative pipeline without opening a SQL file, and it proves the warehouse is live rather than a static extract.

DATA BEHIND THIS TAB

Counts from every MARTS table. Refresh history from INFORMATION_SCHEMA. The seven citations are hand-entered into a small static table, **MARTS.EVIDENCE_CITATIONS** , with columns for claim, figure, issuing body, publication date and URL. Hand-entering them is deliberate: they must never drift.

DATA BEHIND THIS TAB

Counts from every MARTS table. Refresh history from INFORMATION_SCHEMA. The seven citations are hand-entered into a small static table, **MARTS.EVIDENCE_CITATIONS** , with columns for claim, figure, issuing body, publication date and URL. Hand-entering them is deliberate: they must never drift.

TAB 01

Give With Confidence

Two organisations. One is who they say they are. Pick.

TAB 01

Give With Confidence

Two organisations. One is who they say they are. Pick.

TAB 01

Give With Confidence

Two organisations. One is who they say they are. Pick.

S1	4 of 7 viewers picked the imitation this session	
S2	ORGANISATION A	ORGANISATION B
	name	name
	cause · city · state	cause · city · state
	mission line	mission line
	[identical treatment, no tell]	[identical treatment, no tell]
	(I would give to this one)	(I would give to this one)
S3	REVEAL	✓ verified · real filing, good standing
		■ seeded imitation, not real
S4	MEANING VERSUS SPELLING	
	meaning	0.94
	spelling	0.51 ← evasion gap 0.43
S5	CONFIDENCE DECOMPOSITION	S6 SEMANTIC NEIGHBOURHOOD
	waterfall, five components	scatter, imitations sit on their targets
S7	HOW THIS WORKS	1 embed name and mission 2 compare by meaning 3 flag the gap

SEC	ELEMENT	COMPONENT	CONTENT
S1	Hero	hero()	Session accuracy counter. Increments live as the demo runs, which makes the number feel earned rather than printed.
S2	The test	Two comparison cards	Identical typography, identical field order, no visual tell whatsoever before the reveal
S3	Reveal	chip() plus border state	Verified card gains a blue border, imitation gains a red border and a seeded chip
S4	Meaning versus spelling	C01-1	The thesis of the project in one image
S5	Decomposition	C01-2	Waterfall reconstructing the risk score by hand
S6	Neighbourhood	C01-3	Two-dimensional projection of the embedding space

Interaction sequence, exactly

1. Sample one confirmed pair on first load. Randomise which side displays left. Store in session state so a rerun does not reshuffle mid-demo.
2. Both cards render with no tell. Do not colour, badge or shade the seeded one at this stage. Resist the urge.
3. Viewer picks. Record the choice and update the S1 counter.
4. Reveal animates in over roughly 400 ms. This is one of only two permitted animations.
5. Charts S4 to S6 populate for the selected pair.
6. A next pair control resamples without resetting the counter.

C01-1 / PLOTLY GROUPED HORIZONTAL BAR

Meaning versus spelling

Two bars. Semantic similarity in Snowflake Blue, string similarity in Neutral. The visible gap is annotated with an arrow in Flag Red labelled evasion gap. Bars 28 px tall, 6 px radius, axis 0 to 1 with ticks every 0.25. Give this the most vertical space on the tab; it is the single most quotable image in the submission.

C01-2 / PLOTLY WATERFALL

Confidence decomposition

The five components of `F_RISK` building to a total: semantic proximity, evasion gap, geometry flags, unaccounted ratio, missing receipts. Contributions in Flag Red, total bar in ink. An accountability tool cannot show an opaque score, so this chart exists to make the arithmetic reconstructable by eye.

C01-3 / PLOTLY SCATTER, TWO-DIMENSIONAL PROJECTION

Semantic neighbourhood

Roughly 300 organisations in the same cause and region, projected to two dimensions. Verified are Neutral dots, imitations are Flag Red, the selected pair is highlighted with a connecting line and both labels shown. Imitations visibly sit on top of their targets while ordinary organisations spread out.

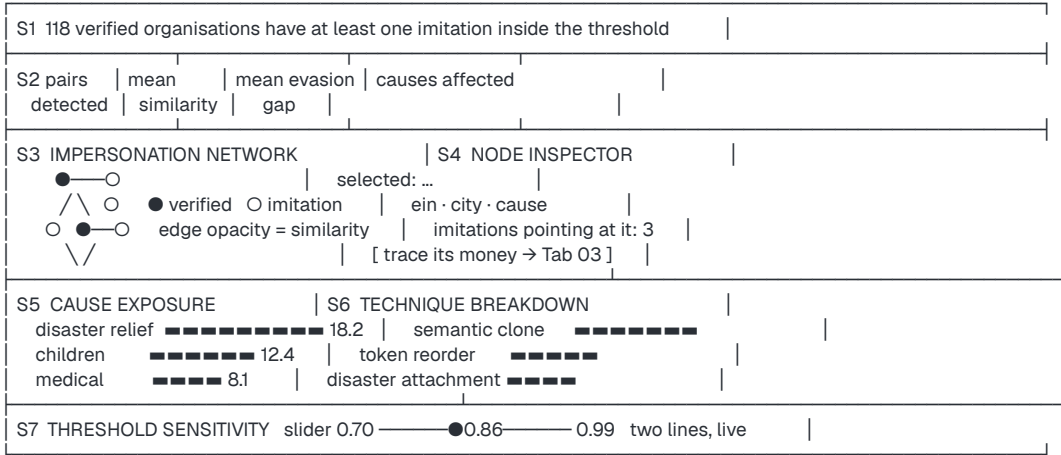
Implementation. Do not attempt UMAP in the warehouse runtime. Compute plain PCA to two components offline and store `pc1` and `pc2` on `MARTS.ORG_PROJECTION`. The tab then runs a simple SELECT and renders instantly.

DATA BEHIND THIS TAB

Real side: IRS Business Master File, unmodified, filtered to organisations in good standing. Suspect side: `tools/make_synthetic.py`, batch `SYNTH_ADVERSARY_V1`. Projection coordinates: offline scikit-learn PCA over the stored vectors. Risk components: `MARTS.ORG_RISK`.

The Trust Graph

One imitation is an incident. Four hundred is a structure.



SEC	ELEMENT	COMPONENT	CONTENT
S1	Hero	hero()	Count of verified organisations with at least one imitation nearby
S2	Stat band	stat_band()	Pairs detected, mean similarity, mean evasion gap, causes affected
S3	Network	C02-1	Force-style graph from plain Plotly traces
S4	Node inspector	Panel bound to selection	Detail for the clicked node plus a control that jumps to Tab 03 carrying focus
S5	Cause exposure	C02-2	Imitations per hundred verified, by cause
S6	Technique breakdown	Horizontal bar	Which generation tactic produced each detected pair. Honest, since the tactics are known by construction, and it usefully shows which techniques the detector handles best.
S7	Threshold sensitivity	C02-3	Interactive slider over a precomputed curve

C02-1 / PLOTLY SCATTER PLUS LINE TRACES

Impersonation network

Third-party network components cannot load under the Content Security Policy, so build the graph from primitives. Compute positions offline with a networkx spring layout and store x and y on `MARTS.GRAPH_NODES`, edges on `MARTS.GRAPH_EDGES`. Verified nodes are large Snowflake Blue, imitations small Flag Red, edges thin Neutral lines with opacity scaled to similarity. Hide axes entirely.

Interaction. Capture Plotly click events through the selection state on `st.plotly_chart` . Clicking writes `focus_org_id` and populates S4.

C02-3 / PLOTLY DUAL LINE DRIVEN BY A SLIDER

Threshold sensitivity

Slider from 0.70 to 0.99. Two lines: pairs detected, and pairs the AI predicate also confirmed. A vertical marker sits at the production value of 0.86. Precompute the entire curve into `MARTS.THRESHOLD_CURVE` so the slider responds instantly with no query.

Why it earns points. It shows the threshold was chosen rather than guessed and lets a sceptical judge probe the model live instead of taking a number on faith.

DATA BEHIND THIS TAB

CLONE_PAIRS and CLONE_CONFIRMED for edges. GRAPH_NODES and GRAPH_EDGES from an offline networkx layout, written once. THRESHOLD_CURVE precomputed by sweeping the cosine cut-off in SQL and storing the counts. Technique labels come from the generator, which records which transformation produced each row.

TAB 03

Follow The Money

Every delivery on a map, coloured by whether the geometry makes sense.

S1 11.4% of traced disbursements have implausible delivery geometry			
S2 hex cells	value traced	outside footprint	impossible transit
S3 H3 HEXAGON MAP, extruded, pitch 45°, height = value, colour = flag rate			
			
S4 DISTANCE FROM BASE histogram, cut-off rule, red tail	S5 VERDICT COMPOSITION donut, four segments, centre = value at risk		
S6 CELL DRILL-DOWN table bound to the selected hexagon, limit 50 rows			
S7 HOW THIS WORKS 1 index to H3 2 measure grid distance from base 3 flag the impossible			

SEC	ELEMENT	COMPONENT	CONTENT
S1	Hero	hero()	Share of disbursements with implausible geometry
S2	Stat band	stat_band()	Cells, value traced, outside footprint, impossible transit

S3	Hex map	C03-1	The signature visual of the whole application
S4	Distance histogram	C03-2	Kilometres from registered base with a plausibility rule
S5	Verdict donut	C03-3	Four-way composition, clickable as a filter
S6	Drill-down	Bound table	Rows for the selected hexagon, hard limit 50
S7	Explainer	step_explainer()	What H3 is, in three sentences, for a judge who has never used it

C03-1 / PYDECK H3HEXAGONLAYER

Delivery geography

H3 cells from `SERVING.V_HEX_RISK` rendered as extruded hexagons. Elevation encodes value delivered. Fill interpolates Snowflake Blue at flag rate zero to Flag Red at flag rate one. Configuration: extruded true, elevation_scale 40, coverage 0.92, pickable true, get_hexagon set to the h3 column, pitch 45. Tooltip styled explicitly in Geist through the pydeck tooltip style dictionary, because tooltips escape the global stylesheet.

Prerequisite. External Offerings Terms must be accepted or the tile layer renders blank with no error. See Trap 04. Fallback if it will not cooperate: a Plotly scatter over latitude and longitude with size and colour encoding, which loses elevation but keeps the geography.

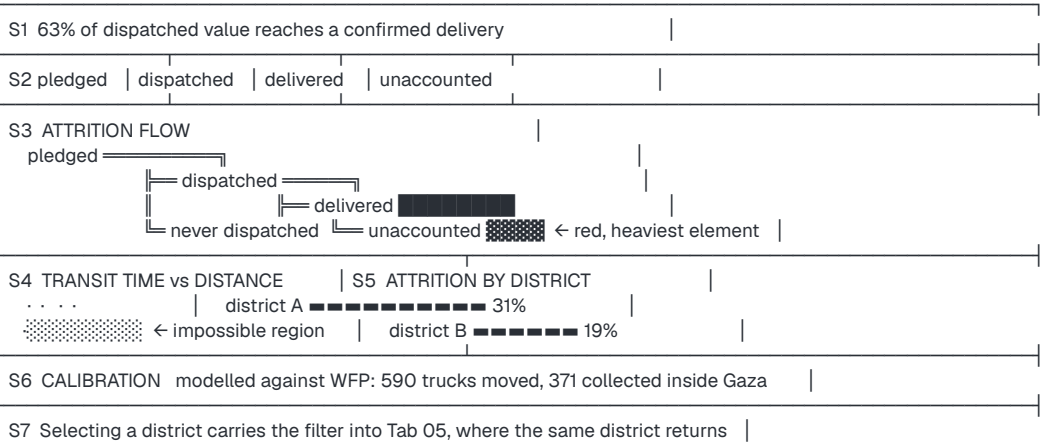
DATA BEHIND THIS TAB

IATI Datastore activity locations provide real geography. Disbursement events are synthetic and labelled. H3 indexing performed in SQL with `H3_POINT_TO_CELL` at resolution 7 for delivery and resolution 5 for organisational footprint. Grid distance via `H3_GRID_DISTANCE`, straight-line distance via `ST_DISTANCE` over `ST_POINT`.

TAB 04

The Last Mile

Money does not vanish at the source. It vanishes between the warehouse and the person.



under privacy protection. Narrate this handoff in the demo.

SEC	ELEMENT	COMPONENT	CONTENT
S1	Hero	hero()	Delivery rate as a single percentage
S2	Stat band	stat_band()	Four stage totals in dollars
S3	Attrition flow	C04-1	Sankey, four node columns
S4	Transit feasibility	C04-2	Scatter with a shaded impossible region
S5	District ranking	C04-3	Bars, clickable, sets the cross-tab filter
S6	Calibration panel	source_note() plus table	States plainly which published figures the synthetic attrition was tuned against
S7	Handoff	Inline note	Explains the deliberate link into Tab 05

C04-1 / PLOTLY SANKEY

Pledged to delivered

Four node columns: pledged, dispatched, delivered, unaccounted. Link width is dollars. The link into unaccounted is Flag Red and is deliberately the visually heaviest element on the tab. Node labels carry both the dollar figure and the share of the original pledge.

C04-2 / PLOTLY SCATTER WITH A FEASIBILITY BOUNDARY

Transit time against distance

X is kilometres, Y is hours in transit. A shaded region below a plausible-speed line marks physically impossible deliveries. Points inside render Flag Red and are individually hoverable with their disbursement identifier. This is the most immediately legible signal in the whole application because it needs no explanation: a truck cannot cover four hundred kilometres in under an hour.

DATA BEHIND THIS TAB

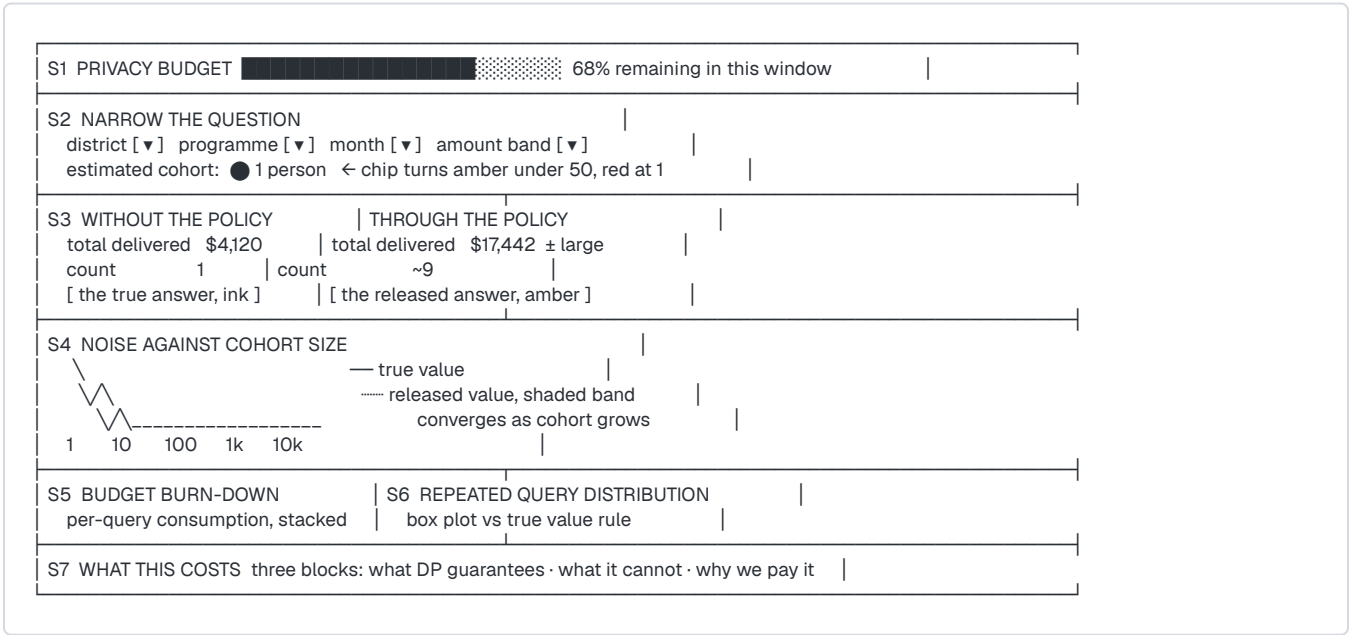
IATI transaction records where available. Delivery events synthetic, with attrition rates tuned so the aggregate delivery share sits close to the published WFP ratio of 371 collected from 590 moved. State this calibration in S6 and again on Tab 10. Do not imply the individual events are real.

TAB 05

The Wall

Now try to use this tool to find one specific person. You will not be able to.

This is the centrepiece. Every tab before it shows the system finding things. This one shows the system refusing to find something, on purpose, with a mathematical guarantee. It is the strongest single screen in the submission and it is never cut.



SEC	ELEMENT	COMPONENT	CONTENT
S1	Hero	hero() plus bar	Live remaining privacy budget, counting down as the viewer queries
S2	Filter console	Four selects plus cohort chip	The attack surface. Chip tone changes as cohort shrinks.
S3	Dual columns	compare_columns()	True answer beside released answer, labelled unambiguously
S4	Noise curve	C05-1	The picture of a privacy guarantee
S5	Budget burn-down	C05-2	Shows a narrow query costs far more budget than a broad one
S6	Repeated queries	C05-3	Defeats the averaging objection
S7	Explainer	step_explainer()	Three honest blocks including what differential privacy cannot do

Interaction sequence, exactly

1. Start broad. Cohort large. Both columns show nearly the same number and the interface says so plainly.
2. Narrow. Below roughly fifty, divergence becomes visible and the cohort chip turns amber.
3. Narrow to one. Chip turns red. The released column either returns a wildly noisy value or the query is refused because the budget is exhausted. Either outcome is a win and the interface explains which occurred.
4. A run this ten times control demonstrates that repeated attempts do not average toward the truth, because the budget depletes.

5. A reset control restores the budget for the next demo run, with an inline note that a production deployment would not expose it.

C05-1 / PLOTLY DUAL LINE, ANIMATED ONCE

Noise against cohort size

X is cohort size, logarithmic, one to ten thousand. True value in ink, released value in Noise Amber with a shaded band showing the plausible range. Lines converge right and diverge violently left. On first render only, sweep a marker right to left over roughly 1.4 seconds so the divergence is felt rather than read. This is the second and final permitted animation.

C05-3 / PLOTLY BOX OR STRIP PLOT

Repeated-query distribution

Spread of released answers across repeated identical queries, against the true value drawn as a vertical rule. Tight around the truth for a large cohort, enormous and centred nowhere useful for a cohort of one.

WHAT TO SAY WHILE DEMONSTRATING THIS

"Every accountability project has the same unsolved problem. To prove aid reached people you have to publish data about those people. This is the part where the tool protects them from me, from you, and from itself." Delivered over the diverging lines of C05-1, that is the moment the submission stops being a hackathon build.

DATA BEHIND THIS TAB

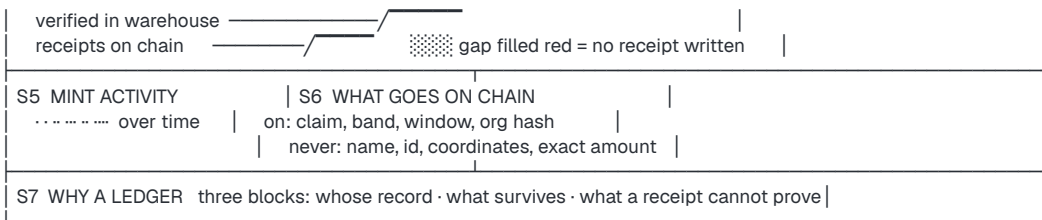
Entirely synthetic beneficiary records. No real personal data enters this project at any point, which is itself the correct engineering decision and should be stated in the interface. The protected object is `SERVING.V_BENEFICIARY_OUTCOMES` with a privacy policy and entity key on `beneficiary_id`. The true column reads the unprotected twin under a privileged role. Both are labelled on screen.

TAB 06

The Receipt

Do not trust this application. Check the ledger yourself.

S1 1,284 receipts on chain		37 expected receipts missing	
S2 tree address	leaves used	capacity remaining	cluster devnet
S3 LOOK IT UP		[paste an asset identifier]	(verify)
RECEIPT FOUND			
programme	AID-2026-DR-114	window	2026-08
amount band	500 to 2K	org hash	3f9a...c17
tree	Bv7...9kQ	leaf index	842
[open in solana explorer]			
S4 EXPECTED AGAINST ACTUAL			



SEC	ELEMENT	COMPONENT	CONTENT
S1	Hero	hero()	Two figures side by side: receipts found and receipts missing. The second number is the interesting one.
S2	Stat band	stat_band() plus identifier()	Tree address, leaves used, capacity remaining, cluster label reading devnet
S3	Lookup and receipt card	identifier() input plus card	Paste an asset identifier, get the decoded receipt with an explorer link
S4	Expected against actual	C06-1	Cumulative step lines with the gap filled red
S5	Mint activity	C06-2	One point per mint over time
S6	Privacy table	Two-column table	What is written on chain and what is deliberately withheld
S7	Explainer	step_explainer()	Whose record it is, what survives if this application disappears, and what a receipt cannot prove

C06-1 / PLOTLY STEP CHART WITH A FILLED GAP

Expected against actual

Two cumulative step lines over time: disbursements marked verified in the warehouse in Snowflake Blue, and receipts confirmed on chain in Solana Purple. Where the lines separate, fill the area Flag Red and annotate it no receipt. The visual argument is that a missing receipt is itself a finding, which is the entire reason a public ledger belongs in this design.

C06-2 / PLOTLY SCATTER OVER TIME

Mint activity

One point per compressed NFT. X is mint time, Y is amount band, colour Solana Purple. Hover reveals the asset identifier and a link. This exists to confirm to a sceptical judge that the volume is genuine rather than three hand-made tokens created for a screenshot.

DEMO SAFETY

Every receipt is minted hours before recording. The lookup reads the local mirror table populated by the bridge, and only the explorer link touches the network. If the RPC is slow during the demo, nothing on screen stalls. Never call an RPC from a render path.

DATA BEHIND THIS TAB

ORACLE.MINT_LOG mirrors every successful mint with asset identifier, signature, tree address, leaf index and explorer URL.

ORACLE.MINT_QUEUE supplies the expected side. **SERVING.V_RECEIPT_GAP** computes the difference over time. The gap is real: a deliberate subset of verified disbursements is left unminted so the finding is genuine rather than staged, and Tab 10 says so.

TAB 07

The Historian

Change a verdict. The warehouse still remembers what it was.

S1 90 days of retention available on this account	
S2 REWRITE THE RECORD organisation [▼] new score [42] (apply) (query the past) demo mode only. a production deployment would not expose this.	
S3 AS IT STANDS NOW risk score 42 verdict verified changed by ACCOUNTADMIN	AS IT STOOD FIVE MINUTES AGO risk score 87 verdict needs a second look [AT(OFFSET => -300)]
S4 VALUE HISTORY 87 42 step function across successive offsets	
S5 CHANGE LOG who · when · from · to	S6 TWO KINDS OF MEMORY warehouse remembers · chain cannot forget
S7 Limits: retention is finite and is controlled by the account owner. The chain is not.	

SEC	ELEMENT	COMPONENT	CONTENT
S1	Hero	hero()	Retention window in days, read from the account rather than hard-coded
S2	Change console	Select, input, two controls	Deliberately lets the viewer tamper, with a demo-mode disclosure
S3	Before and after	compare_columns()	Current row beside the same row at a Time Travel offset
S4	Value history	C07-1	Step function across successive offsets
S5	Change log	Table	Who changed what, when, from which value to which

S6	Two kinds of memory	Side-by-side infographic	Warehouse retention against chain permanence, and the honest difference between them
S7	Limits	Inline note	Retention is finite and controlled by the account owner. Say so; it strengthens the argument for the chain rather than weakening it.

C07-1 / PLOTLY STEP LINE WITH CHANGE MARKERS

Value history

Score for the selected organisation over time as a step function, built by querying successive Time Travel offsets and unioning the results. Each change point annotated with its timestamp. The narration is that an operator can alter the present but not the record of the past, which is the warehouse-side counterpart to the on-chain receipt.

DATA BEHIND THIS TAB

MARTS.ORG_RISK queried with AT(OFFSET) at a series of negative second offsets. No additional storage required, since this is Snowflake internal retention. Enterprise Edition allows up to ninety days, which is why the hero figure reads what it does.

TAB 08

Ask The Warehouse

Accountability that requires SQL is accountability for people who already have power.

S1 1 semantic model · 2 tables · 6 dimensions · 4 metrics · 19 synonyms	
S2 ASK [which causes attract the most imitation?] (ask) (which causes attract the most imitation?) (flagged deliveries by state) (total moved last quarter)	
S3 ANSWER auto-selected chart: bar, line or table ■■■■■■■■■■ disaster relief ■■■■■■■■ children ■■■■ medical	S4 THE SQL IT WROTE SELECT cause, AVG(...) FROM GIVING_SEMANTICS ... never hidden
S5 SEMANTIC MODEL BROWSER facts · dimensions · metrics · synonyms, expandable	
S6 WHY THIS IS AN EQUITY FEATURE, not a novelty. three short blocks.	

SEC	ELEMENT	COMPONENT	CONTENT
S1	Hero	hero()	Shape of the semantic model, counted live from the object rather than typed
S2	Question input	Text field plus three preset chips	Presets guarantee the demo works even if free-form input behaves unpredictably
S3	Answer	Auto-selected chart	

S3	Confidence scatter	C09-1	The warm counterpart to Tab 01's cold open
S4	Leaderboard	Ranked list with inline bars	Top twenty by composite confidence
S5	Detail panel	Reused Tab 02 inspector	Component reuse, not a rebuild
S6	Small multiples	Grid of mini charts	One per cause on shared axes so causes are comparable at a glance
S7	Closing	Prose block	The emotional landing of the entire application

C09-1 / PLOTLY BUBBLE SCATTER

Confidence against delivery rate

X is confirmed delivery rate, Y is receipt coverage, bubble area is value moved, colour by cause using the categorical palette in order. The upper right quadrant is shaded Snowflake Blue at low opacity and labelled give here with confidence. Every dot is a real organisation from the Business Master File.

DATA BEHIND THIS TAB

`SERVING.V_CONFIDENCE_RANK`, built exclusively from IRS Business Master File organisations in good standing, joined to their delivery and receipt statistics. A hard filter excludes any row where `is_synthetic` is true, and an acceptance check verifies that filter holds. This is the only tab in the application with zero synthetic content, and the interface says so in S1.

TAB 10

Method And Honesty

Exactly what is real, exactly what is seeded, exactly what is simplified.

This tab is a scoring instrument. In a project about accountability, a rigorous self-audit is worth more than another feature, and it pre-empts the sharpest objection a judge can raise, which is that the detector finds imitations the builder generated.

S1 1,842,006 real rows · 2,410 seeded rows · every seeded row is labelled in the data			
S2 PROVENANCE, COUNTED LIVE			
batch_id	rows	source	
IRS_BMF_2026	1,802,441	public filings	
IATI_2026	39,565	iati datastore	
SYNTH_ADVERSARY_V1	410	tools/make_synthetic.py	
S3 REAL VERSUS SIMPLIFIED eight-row table, element by element			
S4 WHAT THIS CANNOT DO four honest limitations			
S5 SNOWFLAKE FEATURES USED		S6 REPRODUCE IT	
feature · where · new to me?		numbered sql files, in order	
S7 ETHICS why no real beneficiary data was used, and why devnet not mainnet			

SEC	ELEMENT	COMPONENT	CONTENT
S1	Hero	hero()	Real rows against seeded rows, counted live so the claim cannot drift from reality
S2	Provenance	Live table	Row counts grouped by batch identifier across every table
S3	Real versus simplified	Table	The eight-row table below
S4	Limitations	List	Four honest statements of what the system cannot do
S5	Feature inventory	Table	Every Snowflake feature used, where it appears, and whether it was new to the builder
S6	Reproduce it	Numbered list	The SQL files in execution order, so a judge could rebuild it
S7	Ethics note	Prose	Why no real beneficiary data was used, and why devnet rather than mainnet

ELEMENT	REAL	SEEDED OR SIMPLIFIED
Organisation corpus	IRS Business Master File, unmodified	Nothing
Standing information	IRS Auto-Revocation List, which records failure to file, not wrongdoing	Nothing
Activity and location data	IATI Datastore slice	Coordinates jittered where precision would be identifying
Imitating organisations	Nothing	All of them, generated by tools/ make_synthetic.py, batch SYNTH_ADVERSARY_V1
Beneficiary records	Nothing	Entirely synthetic. Handling real beneficiary data here would be unethical.
Attrition rates	Calibrated against published WFP truck figures	Individual delivery events are modelled
On-chain receipts	Genuinely minted and independently verifiable	Solana devnet, not mainnet
Privacy guarantee	A real Snowflake privacy policy with real noise and a real budget	Epsilon tuned for demonstration legibility rather than production

What this cannot do, S4 text

- It cannot prove intent. A high similarity score is a reason to look closer, never a verdict about a person.
- It cannot detect an imitator whose name shares no meaning with its target. Detection by meaning has a blind spot for names chosen at random.

- It cannot see money that never entered a reporting system at all, which is very likely where the largest losses actually sit.
- A receipt proves a claim was recorded at a point in time. It does not prove goods reached a person. Anyone who tells you a blockchain solves that is selling something.

REUSE THIS TAB

The contents of Tab 10 become the honesty section of the DEV.to post almost verbatim. Write it once, carefully, and use it twice.

08 / Solana bridge

One compressed NFT per verified disbursement. The key never touches the warehouse.

Why a public ledger belongs here

State this argument explicitly in the write-up, because the weakest submissions in any blockchain category are the ones that cannot answer it. A donor verifying a claim about a charity is verifying it against a record kept by the charity, or by a platform the charity pays, or by an application built by someone they have never met. Every one of those requires trust in an interested party. A compressed NFT receipt is a record that persists whether or not GLASSPOCKET exists, whether or not the operator stays honest, and whether or not the warehouse is later edited. That property is the only reason a chain is here, and it is a real property rather than a decorative one.

Design

DECISION	CHOICE	REASON
Cluster	Devnet	Free, sufficient, honestly disclosed in the interface and the post
Token model	Compressed NFT on a Bubblegum Merkle tree	State compression makes per-receipt cost negligible, which is what allows one receipt per disbursement rather than one summary per day
Library version	Bubblegum V1 through Umi	V2 exists but wallet and indexer support is still maturing. For a recorded demo where display must not break, V1 is the lower-risk choice. Note the tradeoff in the post rather than hiding it.
Tree size	maxDepth 14, capacity 16,384	Comfortably exceeds demo volume. Tree parameters are immutable after creation, so oversize deliberately.
Read path	Helius devnet RPC with Digital Asset Standard support	The default public devnet endpoint does not serve DAS
Trust boundary	Node process polls Snowflake, signs locally, writes results back	Trial accounts have no external network access, and a signing key has no business inside a data warehouse regardless

What goes on chain, and what must not

The receipt metadata carries a claim, never a person. Fields: programme code, amount band rather than exact amount, delivery window rather than exact timestamp, a salted hash of the organisation identifier, and a schema version. No beneficiary identifier, no coordinates, no name, no exact figure. The rule to state in the post is: **publish the claim, not the data**. A public ledger is permanent, and permanence plus personal data is a harm that cannot be undone.

Queue contract

```
-- sql/13_oracle_queue.sql
CREATE OR REPLACE TABLE ORACLE.MINT_QUEUE (
  queue_id      STRING DEFAULT UUID_STRING() PRIMARY KEY,
  disbursement_id STRING,
  programme_code STRING,
  amount_band   STRING,    -- '0-500', '500-2K', '2K-10K'
  window_key    STRING,    -- '2026-08'
  org_hash      STRING,    -- SHA2(org_id || salt)
  claimed_at    TIMESTAMP_NTZ,
  status        STRING DEFAULT 'PENDING' -- PENDING, CLAIMED, MINTED, FAILED
);

CREATE OR REPLACE TABLE ORACLE.MINT_LOG (
  disbursement_id STRING, asset_id STRING, signature STRING,
  tree_address STRING, leaf_index INT, minted_at TIMESTAMP_NTZ,
  explorer_url STRING
);
```

Bridge loop

```
// bridge/src/index.ts, shape only
const umi = createUmi(HELIUS_DEVNET_RPC).use(mplBubblegum());
umi.use(keypairIdentity(loadKeypairFromEnv()));

while (running) {
  const batch = await claimBatch(50);    // UPDATE ... SET status='CLAIMED' RETURNING
  for (const row of batch) {
    try {
      const sig = await mintToCollectionV1(umi, {
        leafOwner: TREASURY, merkleTree: TREE, collectionMint: COLLECTION,
        metadata: buildReceiptMetadata(row),
      }).sendAndConfirm(umi, { confirm: { commitment: "finalized" } });

      const assetId = await parseLeafFromMintToCollectionV1Transaction(umi, sig.signature);
      await writeMintLog(row, assetId, sig);
    } catch (e) {
      await markFailed(row, e);           // never crash the loop
    }
  }
  await sleep(4000);
}
```

OPERATIONAL RULES

Confirm at finalized commitment before parsing the asset identifier, or the parse fails intermittently. Poll the DAS API with retries after minting because there is an indexing delay before a new asset is queryable. Airdrop devnet SOL well ahead of time; faucet rate limits are real and will bite on the evening you need them. Run the full mint batch on Saturday afternoon so that Sunday's recording touches nothing live.

09 / Build order

Dependency sorted and risk sorted. Times are indicative, the order is not.

BLOCK	WORK	EXIT CONDITION
Friday evening Hours 1 to 2	Account provisioning. Enterprise edition selected and asserted. Warehouse XSMALL with 60 second suspend. Database and five schemas. External Offerings Terms accepted. Repository scaffolded with the full tree from Section 04. Geist woff2 obtained and geist.css generated.	<code>SELECT CURRENT_EDITION();</code> returns ENTERPRISE. An empty Streamlit app renders with Geist visibly applied and no monospace anywhere.
Friday night Hours 3 to 5	Differential privacy first. Build a minimal beneficiary fact table with fabricated rows, create the privacy policy, attach to a serving view, verify the three behaviours in Section 06.1.	Row-level select fails as GP_ANALYST. Broad aggregate returns with small noise. Single-entity aggregate returns large noise or is refused. If this fails, invoke the Section 10 fallback now, not later.
Saturday morning Hours 6 to 10	Data layer. Load Business Master File and the revocation list. Load an IATI slice. Attach one free Marketplace listing. Generate synthetic adversaries and disbursements. Build staging Dynamic Tables.	Row counts recorded. Every row carries a batch identifier. The synthetic-suspect constraint check passes.
Saturday midday Hours 11 to 14	Intelligence layer. Batch embeddings once. Clone pairs with the mandatory pre-filter. AI_FILTER confirmation on the shortlist. H3 geometry table and hex risk view. Risk UDF. Offline PCA projection and graph node positions written to tables.	Clone pairs populated with a sensible count. Geometry verdicts distributed across all four categories. Nothing in MARTS takes longer than two seconds to query.
Saturday afternoon Hours 15 to 17	Mint everything now. Create the Merkle tree, fund the wallet, run the bridge across the full verified set, populate the mint log. Deliberately leave a subset unminted so the missing-receipt gap on Tab 06 is real rather than staged.	Mint log populated. Three asset identifiers spot-checked in Solana Explorer from a browser.
Saturday evening Hours 18 to 22	Application core. Theme injection, component library, query module. Then Tabs 05, 01 and 06 in that order, because those three alone constitute a complete submission.	The three tabs render, are interactive, and would survive a demo on their own.
Sunday morning Hours 23 to 28	Tabs 03, 04, 02, 00. Map, Sankey, network graph, brief. This is the bulk of the visual payload.	Seven tabs live. Hero figure present on every one.
Sunday afternoon Hours 29 to 33	Tabs 09, 10, 07, 08 in that order of value. Semantic view and Time Travel are the first things sacrificed if the clock is short.	Eleven tabs, or a documented decision about which were cut and why.

Sunday evening Hours 34 to 38	Polish pass against Section 03. Acceptance checks from Section 11. Record the demo video following the Section 12 script. Write the post.	Video recorded. Post drafted. Repository README complete with the honesty table.
Monday morning By 10:00 IST	Publish with the required tag. Final commit. Verify the submission renders correctly on DEV.	Published with more than two hours of margin before the 12:29 IST deadline.

THE SINGLE MOST IMPORTANT LINE IN THIS DOCUMENT

Tabs 05, 01 and 06 are the submission. Everything else is amplification. If at any point the schedule slips, protect those three and cut from the other end. A tight three-tab application with a working privacy guarantee and a real on-chain receipt beats eleven half-finished tabs by a wide margin, and it beats them on the criteria that are actually scored.

10 / Failure modes and cut list

Every known way this build breaks, with a time-boxed response.

FAILURE	DETECT	RESPONSE, TIME-BOXED
Account is not Enterprise, privacy policy DDL rejected	Hour 1	Create a second trial with a different email and select Enterprise. Twenty minutes. Do not proceed on a Standard account.
Privacy policy applies but the Attacker tab queries are rejected for unsupported syntax	Hour 4	Differential privacy supports a subset of SQL. Simplify to COUNT, SUM and AVG with a single GROUP BY. Thirty minutes. If still failing, fall through to the next row.
Differential privacy cannot be made to work at all	Hour 5	Fallback: implement the Attacker tab against an aggregation policy with a minimum group size instead, which enforces a real k-anonymity floor and is still a genuine governance feature. Relabel the tab honestly as a minimum-cohort guarantee rather than a differential-privacy guarantee. Document the attempt and the reason in the post. Sixty minutes.
Dynamic Table refresh fails after a policy is attached	Hour 6	Expected. A privacy-protected object breaks downstream refresh. Move the policy to a terminal view. Ten minutes.
Similarity query runs long or burns credits	Hour 12	The pre-filter has been removed or weakened. Restore region and cause equality in the join. Five minutes.
Cortex daily credit allowance exhausted	Hour 13	Embeddings are already materialised, so nothing user-facing breaks. Skip AI_FILTER confirmation and mark those rows as heuristically confirmed. Disclose in the interface.
pydeck renders an empty canvas	Hour 24	External Offerings Terms not accepted. Accept in Snowsight. Five minutes. If still blank, fall back to a Plotly scatter over latitude and longitude with size and colour encoding, which loses elevation but keeps the geography.
Devnet faucet rate limited	Hour 15	Use an alternative faucet and reduce the tree depth to 12. Fifteen minutes. This is why minting happens Saturday and not Sunday.
Minted assets not yet visible through DAS	Hour 16	Indexing delay. Poll with backoff. The application reads the local mirror table regardless, so the interface is unaffected.
A monospace font appears anywhere	Polish pass	Streamlit's st.code and st.json force monospace. Replace with the custom identifier component. Twenty minutes.

Cut list, strictly in order

- cut first** Tab 08, Ask The Warehouse. Keep the semantic view object, since it still counts as a feature, and remove only the question interface.
- cut second** Tab 07, The Historian. Time Travel is a paragraph in the post if it is not a tab.
- cut third** Tab 02, The Trust Graph. The network layout is the most fiddly visual in the build.
- cut fourth** Tab 04, The Last Mile. Fold the Sankey into Tab 03 if time allows, otherwise drop.

5. **cut fifth** Tab 09, Where A Dollar Lands. Painful, because it carries the emotional close. If cut, move its message into the closing paragraph of the post instead.
6. **never cut** Tabs 00, 01, 05, 06 and 10.

11 / Acceptance criteria

The build is not finished until every box is checked. Run these before recording.

Integrity

- ☐ No real, named organisation appears anywhere in the interface in a flagged or suspicious state. Enforced in SQL, verified by query, not by inspection.
- ☐ Every seeded row carries batch identifier SYNTH_ADVERSARY_V1 and renders a visible seeded chip.
- ☐ The persistent header banner disclosing synthetic data is present on all eleven tabs.
- ☐ Every cited figure in the interface matches the Section 01 table exactly, with its source named.
- ☐ The Auto-Revocation List is described as loss of good standing, never as fraud.

Privacy

- ☐ As GP_ANALYST, a row-level SELECT on the protected view fails.
- ☐ A broad aggregate returns a value close to the truth.
- ☐ A single-entity aggregate is either heavily noised or refused.
- ☐ Repeated identical queries do not converge on the true value.
- ☐ No beneficiary identifier, coordinate or exact amount is written on chain.

Snowflake surface

- ☐ A privacy policy exists and is attached to a serving view.
- ☐ Vector embeddings are stored in a VECTOR column and similarity runs in SQL.
- ☐ At least four distinct H3 or geospatial functions appear in production queries.
- ☐ A semantic view object exists with facts, dimensions, metrics and synonyms.
- ☐ At least six Dynamic Tables refresh within their TARGET_LAG.
- ☐ The risk score is a SQL UDF whose components are displayed in the interface.

Chain

- ☐ More than one hundred compressed NFTs minted on devnet.
- ☐ Three asset identifiers verified independently in Solana Explorer.
- ☐ A genuine gap exists between expected and actual receipts, and the interface explains it.
- ☐ No signing key appears in the repository, in Snowflake, or in the video.

Craft

- ☐ Geist renders on every element. No monospace anywhere, including tooltips, tables and chart axes.
- ☐ Every tab opens with a hero figure above the fold.
- ☐ No default Streamlit chrome is visible.
- ☐ Colour meaning is consistent across all eleven tabs.
- ☐ Every tab renders in under two seconds on an XSMALL warehouse.
- ☐ No AI function, embedding call or unbounded scan executes in a render path.

Submission

- ☐ Repository created and completed inside the contest window, with a commit history that shows it.
- ☐ Post uses the official template with the required weekendchallenge tag.
- ☐ Prize categories section names Snowflake and Solana explicitly.
- ☐ Demo video under three minutes, following the Section 12 script.
- ☐ Honesty table present in both the application and the post.

12 / Copy deck and demo script

Voice: direct, confident, unadorned. Short sentences. No corporate register, no hype vocabulary, no em dashes.

Interface copy

LOCATION	TEXT
Header banner	Real organisation data from public filings. Impersonators and beneficiary records are seeded and labelled. See the Method tab.
Tab 01 prompt	One of these is a real organisation. The other borrowed its meaning. Pick the one you would give to.
Tab 01 reveal, wrong	Most people pick that one. Names are not evidence. Here is what is.
Tab 01 reveal, right	Correct, and it was closer than it looked. Here is the margin.
Tab 05 opening	Everything so far has been this system finding things. Now try to make it find one person.
Tab 05 at cohort one	Cohort of one. The released answer is now noise, and your privacy budget is spent. This is the guarantee working.
Tab 06 opening	You should not have to trust this application. Take an identifier and check it yourself.
Tab 06 missing receipt	No receipt was written for this disbursement. That absence is the finding.
Tab 09 opening	Every organisation on this tab is real and cleared every check. This is what the machinery was for.

Demo video, under three minutes

TIME	SCREEN	NARRATION
0:00 to 0:20	Tab 01, two cards	Open cold on the choice. Ask the viewer to pick before revealing anything. Say that 96 million dollars was reported lost in the United States alone in 2024 to exactly this.
0:20 to 0:45	Tab 01 reveal, C01-1	Reveal. Meaning similarity 0.94, spelling similarity 0.51. The gap is deliberate evasion, and it is computed in SQL inside the warehouse with no data leaving it.
0:45 to 1:05	Tab 03, hex map	Rotate the map once. Point at a red spike. Say that this delivery claims to have landed outside any operating footprint the organisation has.
1:05 to 1:25	Tab 04, Sankey	Show the attrition flow. Cite the WFP figure of 590 trucks moved and 371 collected as the calibration anchor.
1:25 to 2:05	Tab 05, the wall	The centrepiece. Narrow the filters live. Watch the columns diverge and the budget drain. Deliver the line from Section 07 about the tool protecting people from its own operator.

2:05 to 2:30	Tab 06, receipt	Paste an asset identifier, open Solana Explorer in a second window, show it exists independently. Then show a gap where a receipt is missing.
2:30 to 2:50	Tab 09	Close warm. These organisations are real and cleared every check. The point was never to catch people, it was to make giving legible.
2:50 to 3:00	Tab 10	One sentence on what is seeded and what is real. End there.

Post structure

1. **What I Built.** Open with the two-card problem, not with the architecture. Lead with the human situation.
2. **Demo.** Video plus a link.
3. **Code.** Repository embed.
4. **How I Built It.** In headline order: the privacy guarantee, the vector detection, the geography, the receipt ledger. State plainly which Snowflake features were new to you and which you had used before. That self-awareness reads as growth and judges notice it.
5. **What is real and what is simplified.** The Tab 10 table, verbatim.
6. **What this cannot do.** Four honest limitations.
7. **Prize Categories.** Snowflake and Solana, with one paragraph each explaining why the technology is load-bearing rather than decorative.

Title candidates

- A Fake Charity Is a Real One With the Serial Numbers Filed Off
- I Built a Tool That Refuses to Tell You Who a Donation Went To
- Good Intentions Do Not Come With a Receipt, So I Minted Sixteen Thousand

CLOSING INSTRUCTION TO THE IMPLEMENTING AGENT

When the schedule tightens, and it will, protect three things in this order: the honesty of the data labelling, the privacy guarantee on Tab 05, and the working on-chain receipt on Tab 06. Those three are what make this a piece of work rather than another demonstration. Everything else in this document is in service of them.